

THIRUVENKADAPRASATH.S
192312016
EXP20

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression

# Sales dataset
data = {
    "Month": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
    "Sales": [12000, 13500, 15000, 16000, 17500, 19000,
              20500, 22000, 23500, 25000, 27000, 29000]
}

df = pd.DataFrame(data)

# Input and output
X = df[["Month"]]
y = df["Sales"]

# Train model
model = LinearRegression()
model.fit(X, y)

# Predict future sales
future_months = np.array([[13], [14], [15], [16], [17]])

predictions = model.predict(future_months)

# Display results
for month, sales in zip(future_months.flatten(), predictions):
    print("Month",
          month, "Predicted Sales: ₹", round(sales, 2))
```

OUTPUT

Month 13 Predicted Sales: ₹ 29795.45

Month 14 Predicted Sales: ₹ 31302.45

Month 15 Predicted Sales: ₹ 32809.44

Month 16 Predicted Sales: ₹ 34316.43

Month 17 Predicted Sales: ₹ 35823.43s